



OASIS-3: IMAGING METHODS AND DATA DICTIONARY

VERSION 2.3 JUNE 2025



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INTRODUCTION & CONTACT INFORMATION

OASIS-3 is the latest release in the *Open Access Series of Imaging Studies (OASIS)* that aimed at making neuroimaging datasets freely available to the scientific community. By compiling and freely distributing this multi-modal dataset, we hope to facilitate future discoveries in basic and clinical neuroscience. Previously released data for OASIS-Cross-sectional (Marcus et al, 2007) and OASIS-Longitudinal (Marcus et al, 2010) have been utilized for hypothesis driven data analyses, development of neuroanatomical atlases, and development of segmentation algorithms. OASIS-3 is a longitudinal neuroimaging, clinical, cognitive, and biomarker dataset for normal aging and Alzheimer's Disease.

The OASIS datasets provide the community with open access to a significant database of neuroimaging and processed imaging data across a broad demographic, cognitive, and genetic spectrum an easily accessible platform for use in neuroimaging, clinical, and cognitive research on normal aging and cognitive decline.

OASIS-3 is a retrospective compilation of data for 1378 participants that were collected across several ongoing projects through the WUSTL Knight ADRC over the course of 30 years. Participants include 755 cognitively normal adults and 622 individuals at various stages of cognitive decline ranging in age from 42-95 yrs. All participants were assigned a new random identifier and all dates were removed and normalized to reflect days from entry into study. The dataset contains 2838 MR sessions which include T1w, T2w, FLAIR, ASL, SWI, time of flight, resting-state BOLD, and DTI sequences. PET imaging from different tracers, PIB, AV45, and FDG, totaling over 2154 raw imaging scans are also available in OASIS-3. Additionally, 449 Tau PET sessions are now available for OASIS-3 subjects in a subproject 'OASIS-3_AV1451', as well as an additional longitudinal set of 81 Tau PET sessions in a subproject called 'OASIS-3_AV1451L'.

AVID CONTACT INFO FOR FLORBETAPIR (AV45) PRESENTATIONS AND PUBLICATIONS

All proposed publications or presentations using Florbetapir F18 (AV45) or Flortaucipir F18 (AV1451) PET data must be submitted to Avid Radiopharmaceuticals for review and comment 30 days prior to such presentation or publication for review of intellectual property interests. Please reference OASIS-brains.org and Washington University (Drs. Morris and Benzinger) in your email.

- Contacts:
 - o Patricia Aldea Stevenson <stevenson@lilly.com>

OASIS PROJECTS AND CITATIONS

Each OASIS project should be used independently and not combined. Due to anonymization participants may be included in all three datasets (OASIS-1, OASIS-2, and OASIS-3) under unique IDs.

- OASIS-1: Cross-Sectional T1w MR images across the lifespan (ages 18-96) with dementia status (doi: [10.1162/jocn.2007.19.9.1498](https://doi.org/10.1162/jocn.2007.19.9.1498))
- OASIS-2: Longitudinal T1w MR images in older adults (ages 60-96) with dementia status (doi: [10.1162/jocn.2009.21407](https://doi.org/10.1162/jocn.2009.21407))

- OASIS-3: Longitudinal Multimodal Neuroimaging (ages 42-95) with dementia status ([doi.org: 10.1101/2019.12.13.19014902](https://doi.org/10.1101/2019.12.13.19014902))
- OASIS-4: Clinical Cohort: <https://doi.org/10.1016/j.nicl.2020.102248>

DATA RELEASES

RELEASE 1.0: MARCH 2018

Initial Longitudinal release of data

- 1100 Subjects (age 42-95)
- Neuroimaging:
 - 2118 MR Subjects
- Clinical and Cognitive Measures:
 - 6534 Longitudinal Clinical follow-up assessments
 - Neuropsych Assessments
 - 4089 NACC UDS2 Assessments

RELEASE 1.2: APRIL 2018

- 1435 PET Sessions

RELEASE 2.0: JULY 2022

Updated longitudinal data to existing subjects and addition of new subjects.

- **New subjects and longitudinal session**
 - 276 Subjects (age 42-100)
 - Neuroimaging:
 - 655 MR Sessions
 - 1472 CT sessions
 - 543 PET sessions
- **Spreadsheets for download without using xnat tables**
- **UDS v2 and UDS v3**
- **Cohort Replication**
 - Option for storage of cohort data to replicate published results
- **OASIS_AV1451**
 - Sub-project with Tau imaging on OASIS3 participants
- **OASIS_AV1451L**
 - Sub-project with longitudinal Tau data on OASIS3 participants

CHANGE LOG RECORDS

- This is a changelog to track changes to the OASIS database.
 - https://github.com/NrgXnat/oasis-scripts/blob/master/OASIS_CHANGELOG.md

DEMOGRAPHICS

TABLE 1-2. SUBJECT DEMOGRAPHICS

Table 1. Subject Demographics

	N	AGE	Right Handed
F	756	68.15 (43.2-95.6)	684
M	622	70.16 (42.5-91.7)	551
Total	1378	69.06 (42.5-95.6)	1235

Table 2. OASIS-3 Clinical Dementia Rating (CDR) Summary

	<i>max CDR</i>					
<i>min CDR</i>	0	0.5	1	2	3	Grand Total
0	755*	215	61	38	7	1076
0.5		83	57	74	7	221
>1			33	36	5	74
Grand Total	755	298	151	154	19	1377

**Unchanged CDR = 0 represents cognitively healthy population. List provided on Central*

OASIS FILE DESCRIPTION

BIDS FILE SPECIFICATION

All MR and PET imaging files are converted to nifti format utilizing the BIDS format (Gorgolewski et al., 2016). This allows for standardized naming and file formats. Raw MR files, in DICOM or IMA format were converted to Nifti format using dcm2nii (DICOM=dcm2niix v1.0.20171017 and IMA=dcm2nii mricronlx64-2013.06.12; Li et al., 2016). In addition to nifti files, a supplemental json file is included with additional acquisition header information, such as TR, TE, flip angle, and scanner model, that is absent from nifti headers.

Documentation on BIDS can be found here (<http://bids.neuroimaging.io/>).

MR IMAGES

- anat
 - T1w
 - T2w
 - TSE (acq-TSE_T2w)
 - T2star
 - FLASH
 - Flair
 - Time of Flight (acq-TOF_angio)
- func
 - Task-rest_bold
 - ASL
- fmap
 - Fieldmap
- dwi
 - DWI
- swi
 - Magnitude (part-Mag_GRE)
 - Phase (part-Phase_GRE)
 - Minimum Intensity Projection (minIP)
 - SWI

PET IMAGES

- pet
 - acq-PIB_pet
 - acq-AV45_pet
 - scq-FDG_pet
 - acq-AV1451_pet* (*only in project: OASIS3_AV1451)

CT IMAGES

- ct
 - CT

AUXILIARY & DERIVATIVE FILES

The dataset includes 2 zip files: one called “OASIS3_auxiliary_files.zip” containing auxiliary files and a second called “OASIS3_derivative_files.zip” containing derivative files. The auxiliary zip contains:

- supplemental BIDS .json files ([described above](#))
- .bvec (vector table) and .bval (vector of b-values) files for the DWI scans
- .tsv timing files for PET images describing the length of each time frame

The auxiliary zip file is initially separated into the main OASIS3 dataset, the OASIS3 AV-1451 dataset, and the OASIS3 AV-1451 Longitudinal dataset. These are then separated by modality, then session, and finally image type. Each image gets its own directory with an Image ID, so if there are multiple scans of the same type in a session, they will each be given a different number (for example, a T1w, a T2w, and a T2star would be given the IDs anat1, anat2, and anat3). The filenames for the auxiliary files, as well as the information in the BIDS .json file, will make it clear which scan the files describe. This structure is illustrated below:

- Dataset (OASIS3, OASIS3_AV1451, or OASIS3_AV1451L)
 - Modality (MR, PET, or CT)
 - Session label (OAS<OASIS Subject ID>_<Modality/Tracer>_d<Days From Entry>, ex: OAS30112_MR_d0098)
 - Image ID (pet1, swi3, anat2, func1, etc.)
 - Auxiliary files (BIDS .json, .bval/.bvec for DWIs, .tsv timing file for PETs)

The derivatives zip contains “pmap” .nii.gz files, which contain TrUE-Net (Vaathi et al, 2021) probability segmentation (v1.0.1) of White Matter Hyperintensities for the MR sessions. Voxel values contain the probability of a given voxel to be WMH. These are all in a single folder called “truenet_oasis3”, with filenames that identify which session they were calculated on. These filenames are in the format OAS<OASIS Subject ID>_MR_d<Days From Entry>_<Image ID>_pmap (ex: OAS30001_MR_d2430_anat3_pmap.nii.gz)

DTI IMAGING

The OASIS-3 dataset includes 1205 DTI. All sequences include a .bvec and .bval file that includes information on the vectors and b-values as estimated through the dcm2niix conversion. These files are found in the auxiliary zip file described above. DTI sequences collected on Siemens scanners are known to have a variance of +/- 10%. Below is the standard vector table for Siemens 25-direction DTI.

VECTOR TABLE FOR 25 DIRECTIONS

Normalisation = none

25 direction

0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24 -0.200000,-0.457663,-0.619678,-0.647200,-0.529196,-0.163313,-0.305531,-
0.346410,-0.294225,-0.334708,-0.876353,-0.107041,-0.174797,-
0.222823,0.000000,0.000000,0.000000,0.000000,0.231234,0.435890,0.489898,0.223607,0.365180,0.626649,0.723592,0.809004
0.000000,0.000000,0.000000,-0.420560,-0.529196,0.163313,0.305531,0.589382,-0.770361,0,-0.294691,0.000000,0.000000,-0.446071,-
0.815963,-0.142720,0.214080,-0.636606,-0.458295,0.515079,0.380445,0.365180,-0.407205,0.000000,0.000000 0,-
0.174796,0.236674,0.21032,-0.529196,-0.163313,0.305531,0.112583,0,0.147328,-0.538023,0.685848,0.721758,-0.504234,-0.37368,-
0.56052,0.318265,-0.599959,0.674296,0.072672,-0.365180,0.203641,-0.525744,0.58780

TABLES 3-6. OASIS-3 IMAGING SUMMARY

Table 3. OASIS-3 MR and PET Imaging Summary

OASIS-3 – Longitudinal Normal Aging and Alzheimer Disease								
	Subjects	MR	CT	PET	AV45	PIB	FDG	AV1451
2018 Release	1100	2183	0	1611	492	1002	117	
2022 Release	276	655	1472	543	287	246	10	
Total OASIS-3	1376	2838	1472	2154	779	1248	127	
OASIS3_AV1451 (restricted access)	449							449
OASIS3_AV1451_longitudinal (restricted access)	80							81

Table 4. OASIS-3 MR Scan Type Inventory

Scan Type	T1w	T2w	FLAIR	Bold – Resting State	DTI	ASL	SWI	TOF	Fieldmap
OASIS-3									
1.5T MR 3.0T MR (Release 2018) 3.0T MR (Release 2022)	654	455	0	0	0	0	0	0	0
	2757	3555	757	3665	2465	1128	1224	673	2926
	695	1029	621	1449	798	432	6	220	890
OASIS-3 – Longitudinal Normal Aging and Alzheimer Disease									
PIB AV45 FDG Total	Release 2018			Release 2022					
	HR+	PET/CT	PET/MR	HR+	PET/CT	PET/MR	Total		
	411	580	8	11	239		1249		
	1	23	467	2		288	781		
	117			10			127		
	529	603	475	23	239	288			

FIGURE 1 LONGITUDINAL IMAGING

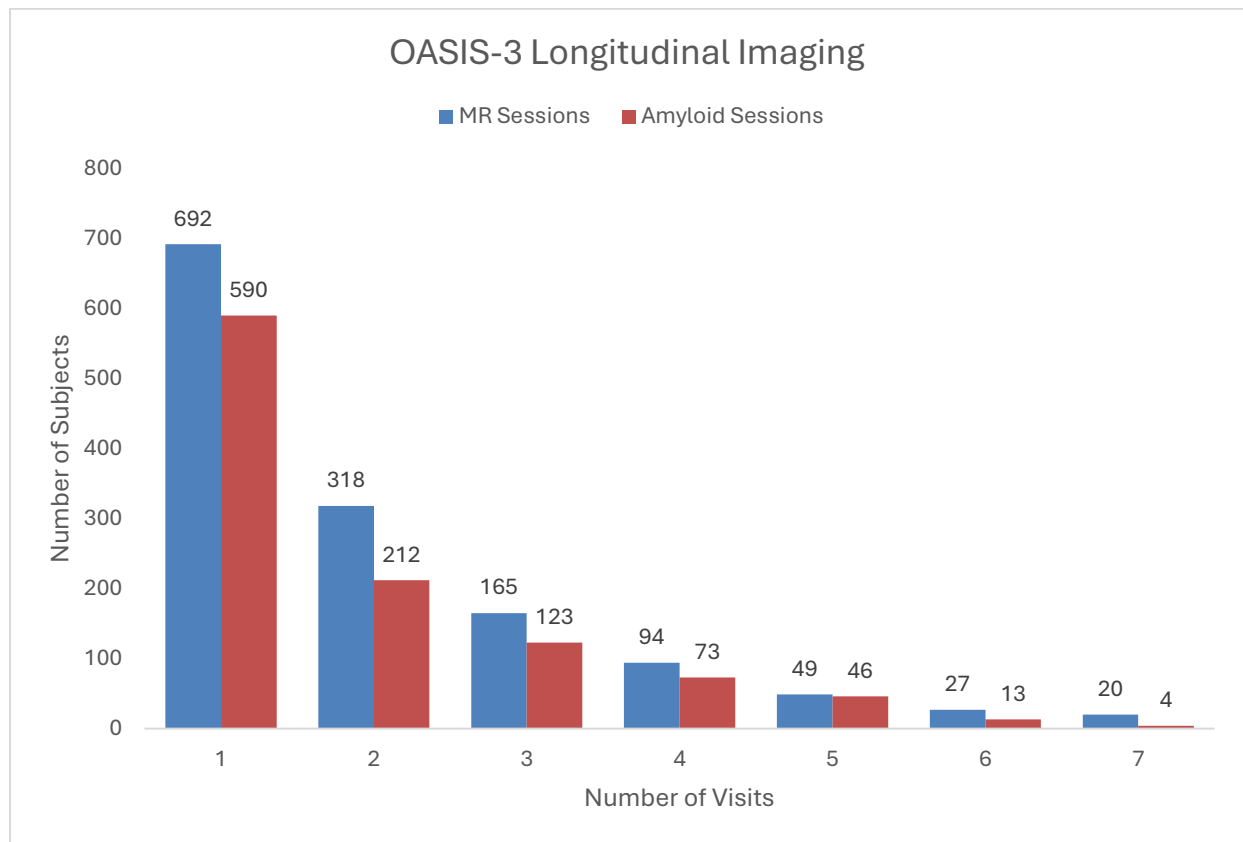


Figure 1. Longitudinal Imaging

MR IMAGING

MR SCANNERS

Data included in OASIS was collected on the following scanners. Scanner specific information is recorded in dataset_description.json for each MR scan session. *For manuscripts, select only the scanner(s) from which your subset of data were derived.*

- Siemens BioGraph mMR PET-MR 3T scanner (51010)
- Siemens TIM Trio 3T MR scanner (2 scanners) (35177 & 35248)
- Siemens Magnetom Vida 3T MR scanner (175614)
- Siemens Prisma_fit 3T MRI scanner (167047)
- Siemens Sonata 1.5T scanner (21926)
- Siemens Vision 1.5T scanner
- Siemens Avanto 1.5T scanner (26321)

SCANNING METHODS

- Participants were placed in the scanner head to foot while lying in the supine position.
- Head immobilization was done by placing small foam cushions between the head and the head coil.
- In many participants, a Vitamin E capsule was used to mark the left temple.
- For all scans a 16-channel head coil was used.
- Participants receiving simultaneous PET acquisition, on the BioGraph mMR were injected with tracer prior to initiation of MRI scanning
- Magnetom Vida scan sessions were converted to NIFTI format from enhanced DICOM format
- *Note that **not** all subjects will have every type of image data.*

RESTING STATE SCANS – SLICE TIMING

Resting state scans are labeled according to BIDS standard “task-rest”. Slice timing is included in the JSON when available. Some original DICOM files did not have slice timing info. **When slice timing information is unavailable use the ratio TR/#slices.**

ASL PARAMETERS

The following parameters are important for processing ASL images but were not extracted into the JSON files as part of the BIDS conversion.

Table 7. ASL Parameters by scanner

Scanner model	Software Version	TI1	TI2	Labeling Strategy	Slice Readout	Background Suppression
51010	syngo_MR_B18P	700	1900	PASL	46.0ms	None
51010	syngo_MR_B20P	700	1900	PASL	46.0ms	None
35177	syngo_MR_B17P	700	1900	PASL	36.4ms	None
35248	syngo_MR_B17P	700	1800	PASL	36.4ms	None
175614				PASL		

PET IMAGING

PET SCANNERS

Data included in OASIS was collected on the following scanners. Scanner specific information is recorded in dataset_description.json for each MR scan session. *For manuscripts, select only the scanner(s) from which your subset of data were derived.*

- Siemens Biograph mMR PET-MR 3T scanner (serial#: 51010)
- Siemens Biograph 40 PET/CT scanner (serial#: 1003)
- Siemens Biograph 128_Vision Edge PET/CT (serial#: 11009)
- Siemens ECAT HRplus 962 PET scanner (serial#: 101006)

HR+ IMAGE NOTES (PIB AND FDG)

ECAT PET images were acquired on either a Siemens 962 HR+ PET scanner. Emission data were reconstructed using a 3-D filter back-projection into a 128 x 128 x 63 matrix (2.12 x 2.12 x 2.43mm) with single scatter correction [1], a ramp filter (resolution of ~5.5-6 mm full-width half maximum [FWHM]), measured attenuation factors from a Ge68 transmission scan, and randoms correction. These images have been converted from ECAT to NIFTI using dcm2niix (v1.0.20180404 GCC4.4.7).

TRACERS

PIB

N-methyl-^[11C]2-(4'-methylaminophenyl)-6-hydroxybenzothiazole (^[11C]PIB) is a radiolabeled compound that binds *in vivo* to brain amyloid deposits. Developed at the University of Pittsburgh, PIB has very high affinity for amyloid plaques. With administration of 6 - 20 mCi of ^[11C]PIB, a 60 minute dynamic PET scan in 3D mode (septa retracted) will be initiated (24 x 5 sec frames; 9 x 20 sec frames; 10 x 1 min frames; 9 x 5 min frames).

AV45

Florbetapir binds to β -amyloid (A β) plaque utilizing the radioactive isotope 18F for use in PET scanning. Florbetapir F18 is used under the research number 18F-AV-45 and therefore referred to as AV45. Participants received a single i.v. administration of approximately 370 MBq (10 mCi) of florbetapir F 18 (over 10-60 sec). There are two acceptable procedures for obtaining the florbetapir F 18 PET scans:

1. In the preferred approach, the participant will be positioned in the PET-MR scanner at the time of injection and a 70-minute dynamic scan (with simultaneous PET and full Standard MR acquisition) will be obtained starting at the time of injection. For florbetapir F 18 scans conducted on the PET/MR scanner, a short (approximately 15 minute) CT scan may be conducted on the PET/CT scanner.
2. For those participants who cannot tolerate the full exam, an alternative is to rest quietly in an uptake room for the first 40 minutes after injection. The participant will then be positioned in the PET-MR scanner to undergo a scan lasting 20 minutes, beginning 50 minutes after florbetapir F 18 injection and lasting for 20 minutes, using the Short MR Protocol. For florbetapir F 18 scans conducted on the PET/MR scanner, a short (approximately 15 minute) CT scan may be conducted on the PET/CT scanner.

FDG

Metabolic imaging with ^[18F]FDG-PET was performed with a 3D dynamic acquisition began 40 minutes after a bolus injection of approximately 5 mCi of FDG and lasted for 20 minutes.

AV1451

Flortaucipir binds to the paired helical filaments characteristic of neurofibrillary tau tangles in Alzheimer disease. Flortaucipir F18 is used under the research name 18F-AV-1451 and therefore referred to as AV1451. Participants received a single i.v. administration of approximately 370 MBq (10 mCi) of flortaucipir F18 (over 10-60 sec). There are three acceptable procedures for obtaining flortaucipir F18 PET scans:

1. In the preferred approach, the participant will be positioned in the PET/CT scanner at time of injection and a 105-minute dynamic scan will be acquired starting at time of injection.
2. For participants who cannot tolerate the full exam, an alternative is to rest quietly in an uptake room for the first 75-minutes after injection. The participant will then be positioned in the PET/CT to undergo a scan lasting 30 minutes.
3. A final alternative is a hybrid approach. In this method, the participant will be positioned in the PET/CT scanner at time of injection and a 60-minute dynamic scan will be acquired starting at time of injection. The participant will then have a 15-minute rest period where they are free to leave the scanner. Following this 15-minute break period, the participant will be placed back into the PET/CT and a 30-minute static scan will be acquired starting at 75-minutes post-injection. The two scans will be reconstructed separately from one another. This method accommodates participants who are unable to complete the full, uninterrupted exam while persevering early frame dynamic information.

REFERENCES

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